

Regular Expressions

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Topics to cover

- what are regular expressions?
- types of regular expressions
- ranges
- quantifiers
- pattern modifiers
- metacharacters
- some examples

What are regular expressions?

- filter for text processing
- the expression itself specifies a text string
- can be quite detailed - very powerful
- used in several scripting languages - perl, python, shell scripts, sed, awk, and a few windows scripting languages
- useful for a range of things to do from simple text matching with grep to a full-fledged natural language processing tool with perl and REs

Types of Regular Expressions

- there are two types of regular expressions:
 - basic regular expressions (BRE)
 - extended regular expressions (ERE)
- most tools use BREs, but some use EREs

Type	grep	sed/ed	ex/vi	egrep	perl	python	awk
BRE	•	•	•				
ERE				•	•	•	•

Ranges

.	any character but newline (\n)
[abc]	matches a or b or c
[a-zA-Z]	any character, case insensitive
[1-9]	any number between 1 and 9
(abc def)	pattern "abc" or "def"
[^xyz]	not x or y or z
^abc	match abc at the beginning of a line
abc\$	match abc at the end of a line
\n	nth subpattern

Quantifiers

*	0 or more
+	1 or more
?	0 or 1
{4}	exactly 4
{2,}	4 or more
{1,5}	between 1 and 5

Pattern modifiers

g	do a global match
i	case-insensitive match
m	match on multiple lines
s	treat the entire text as one line
x	don't ignore comments and white space
e	evaluate replacement

Metacharacters

- Metacharacters are characters that are interpreted as a command or thing to do in a regular expression
- these must be "escaped" - we tell the RE parser to treat them as actual characters
- these are:

^	*	+	{
.	(	)	?
<	>		\
[	\$		

- to escape a metacharacter, we put a "\" before it
- e.g. \. to escape a .

Some examples

- use with grep
- use with sed
- random strings without special characters using REs and sed
- perl-specific RE parts
- word-boundary to pick out words

RE Example 1 - grep

- some examples of filters for searching in files:

```
grep -i "junk" myjunk.txt
```

```
grep -E '(this|is)' myjunk.txt
```

```
grep -i "[Jj]unk" myjunk.txt
```

```
grep -i "[Aa][Bb][Cc]" *.txt
```

- only basic REs in shell, need egrep or "grep -E" to get extended REs
- some examples with grep -E:

```
grep -E '(*eq|[Ll][Ii]*)' $SCRDIR/*.sh  
grep "SCF Done" myg09job.log  
ls -rlth /usr/local | grep -i "cpmd"
```

RE Example 2 - sed

- examples using sed (stream editor):

```
sed s/"NC "/" /g tempfile
```

```
sed -e s/"0-0"/"0 -0"/g \
```

```
-e s/"1-0"/"1 -0"/g \
```

```
-e s/"2-0"/"2 -0"/g \
```

```
-e s/"3-0"/"3 -0"/g \
```

```
-e s/"4-0"/"4 -0"/g \
```

```
-e s/"5-0"/"5 -0"/g \
```

```
-e s/"6-0"/"6 -0"/g \
```

```
-e s/"7-0"/"7 -0"/g \
```

```
-e s/"8-0"/"8 -0"/g \
```

```
-e s/"9-0"/"9 -0"/g $1 > temp.dos
```


RE Example 3 - random char strings

- more complex example with sed:

```
scratch="`cat /dev/urandom | \
    uuencode -m /dev/stdout | \
    head -2 | \
    sed -n -e '2s/^\(.....\).*$/\1/p' | \
    sed -e 's/\/_/g' -e 's/\\V-/g' -e 's/&/=/g' | \
    sed -e "s/^'/X/g" -e "s/^\\`/Y/g" -e "s/^\\\"/Z/g"`"
```

- result is (this run - changes each time):

wf_47EJB9FRI1K29X

- useful for generating random strings of characters
- sed finds special characters according to the above REs and replaces them with other characters

RE example 4 - EREs in perl

- perl has a few handy pattern matching things that are very useful:
 - `\b..\b`: search on word boundary
 - `\B...\B`: search on all but word boundary
 - `\w`: matches a "word" character (alphanumeric and `_`)
 - `\W`: non-word character
 - `\s`: matches whitespace character
 - `\S`: matches non-whitespace character
 - `\d`: matches digit
 - `\D`: matches non-digit

RE example 4 - word boundary in perl

- useful example for REs in perl using word boundaries and alphanumeric characters:
 - pick the words out of a sentence

```
if($line =~ \bword\b) {  
    print "found word in $line";  
}
```
 - more advanced - use to pick out specific words from text or to remove non-alphanumeric characters
 - modify the above example to pick out specific words

More information

- man pages for some tools
- the internet - tutorials, documentation a-plenty!
- "Mastering Regular Expressions", an O'Reilly book
- "Regular Expressions Cookbook", another O'Reilly book - good for perl and python
- local CRC
- support@ace-net.ca